live in places that can't be reached by fibre optic cables due to economic viability. The entrepreneurially inclined audience will perceive this major problem as a major business opportunity. Two-thirds of the world without internet is potentially double the number of internet subscribers that exist today as potential subscribers. The crucial hurdle is making sure that the cost to the subscriber is low enough that it makes sense to subscribe in the first place. It will benefit both sides as the Internet is the gateway to a level playing field for the whole world, and a gateway to much needed progress and development for many. With the ultimate goal of affordable Internet access for all, who will rise to the challenge?

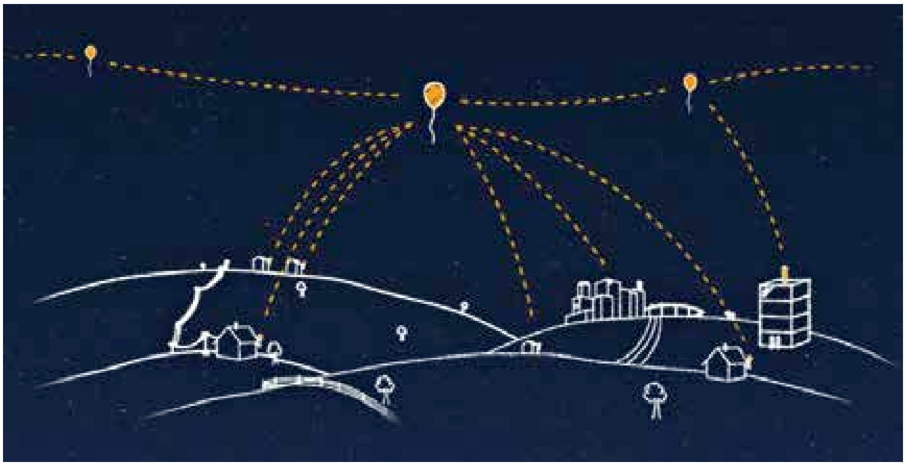
The Game: Internet for All

There are currently few technology-industry leaders competing for this goal with dedicated projects to deploy novel infrastructure. However, solutions for obtaining connectivity around the globe have existed for almost 15 years now in the form of satellite constellations, which are unfortunately not commercially viable without targeted business plans, and abandoned TV broadcasting frequencies

Google was the first company to announce a global-scale internet-for-all project with much hype and media attention. In June 2013, Google Loon was revealed, a moonshot that had been in development within Google X labs as early as 2011. Their plan to solve the internet coverage problem? Balloons! No, this is not another imagination exercise. Fifteen metre wide Loon balloons will ride stratospheric wind currents 18-25 km above the surface of the Earth, well above airplanes and well below satellites. A network of these Loon balloons will synchronise and coordinate their movement to cover and provide connectivity to the required area, with each individual Loon balloon covering an area 40km in diameter. The balloons are filled with Helium and have a variable buoyancy system in order to control their altitude, which determines which wind current they are riding in the Stratosphere, and therefore the direction of movement. The balloons each last between 100-200 days in the atmosphere, although the electronics are re-used after it is taken out of service. The electronics hang under the balloon, and get their power from optimally aligned single-crystal high efficiency solar panels. They have a Lithium-ion battery pack for storing the energy required for use at night. The Loon balloons have transceivers that communicate on the freely available unlicensed ISM (Industrial, Scientific, Medical) bands of 2.4 GHz and 5.8 GHz, and LTE. Users have to install a special antenna to be able to connect to the internet from their remotely located homes. Information travels through the antenna, a chain of balloons, and finally to a ground station that has a connection to an ISP.

The Google's Loon project doesn't aim to provide standalone internet connections but utilises preexisting telecommunications infrastructure and extends it via their balloon network. Loon was first deployed and tested in New Zealand, where a farmer was the first person to use the service. Previously, his only means of connection was a satellite connection which cost over a thousand dollars! In addition to the antenna, users can get coverage directly on the phones and other LTE enabled devices (dongles) using a special Loon SIM card. After the New Zealand tests, the project has been being evaluated in Brazil, and most recently Google has signed a contract with the government of Sri Lanka. It will be the first country to have complete Loon supplied internet coverage in the world!

Soon after the announcement of Loon, in August 2013, Facebook, Samsung, Ericsson, Nokia, Opera Networks, Qualcomm and MediaTek announced the creation of internet.org. It's a non-profit organisation aiming to deliver affordable access to selected internet services in developing countries. Not long after launch, the consortium had been criticised for being too selective with the access provided. In other words, for violating net neutrality. The consortium have opened up since then, creating the internet.org platform for anyone who meets certain specified conditions. It also launched an as-yet-unnamed project, whose official description hasn't revealed more than the fact that it will be a network of UAVs called Aquila drones, which were developed by engineers from Ascenta – an aerospace company that Facebook acquired for this purpose. The drones beam down Internet connectivity using high-throughput



The Google Loon project employs high-altitude balloons to provide internet connectivity

The Google Loon visualised